

# Auditing Model Substitution in LLM APIs

Why software tests fail and hardware attestation wins

Based on: [arXiv:2504.04715](https://arxiv.org/abs/2504.04715) (April 2025)

# The Model Substitution Problem: Economic Incentives Drive Covert Swaps

- Black-box LLM APIs provide no guarantee the advertised model is served.
- Providers have strong incentives to reduce GPU cost while maintaining pricing.
- Formalization: hypothesis test over output distributions  $P_{\text{spec}}(y|x)$  vs  $P_{\text{actual}}(y|x)$ .

## Substitution types (examples)

- (1) Smaller model (e.g., 8B instead of 70B)
- (2) Quantized variant (FP8/INT8 instead of FP16)

- (3) Fine-tuned / updated version
- (4) Different model family / architecture

## Hypothesis test

$H_0$ :  $\text{API} \sim P_{\text{spec}}(\cdot|x)$

$H_1$ :  $\text{API} \sim P_{\text{actual}}(\cdot|x)$

Goal: distinguish with finite queries

# FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

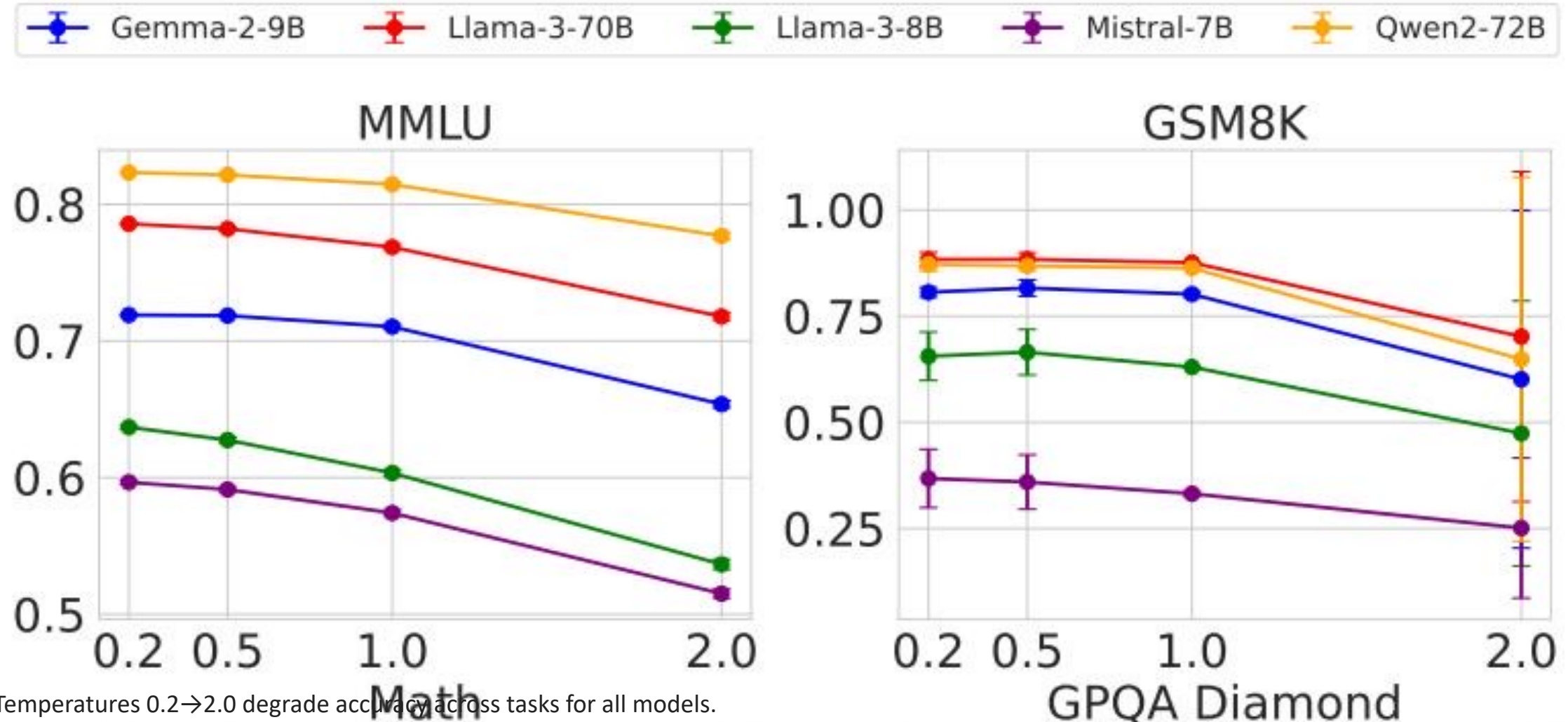
| Model (original vs FP8)      | MMLU         | GSM8K        | MATH         | GPQA Diamond |
|------------------------------|--------------|--------------|--------------|--------------|
| Llama-3-8B-Instruct          | 62.69 ± 0.18 | 61.14 ± 3.47 | 20.65 ± 3.43 | 22.62 ± 0.22 |
| Llama-3-8B-Instruct-FP8      | 62.43 ± 0.26 | 60.90 ± 4.10 | 14.91 ± 2.49 | 20.14 ± 0.24 |
| Llama-3-70B-Instruct         | 78.05 ± 0.08 | 88.06 ± 1.44 | 35.69 ± 1.33 | 29.60 ± 0.51 |
| Llama-3-70B-Instruct-FP8     | 77.88 ± 0.13 | 87.35 ± 1.24 | 35.75 ± 1.16 | 33.12 ± 0.30 |
| Gemma-2-9b-it                | 71.86 ± 0.08 | 81.80 ± 1.35 | 33.41 ± 0.28 | 28.64 ± 2.97 |
| Gemma-2-9b-it-FP8            | 71.92 ± 0.11 | 79.41 ± 1.14 | 32.53 ± 0.34 | 27.81 ± 3.22 |
| Qwen2-72B-Instruct           | 82.18 ± 0.08 | 86.72 ± 1.00 | 37.39 ± 1.41 | 29.93 ± 2.71 |
| Qwen2-72B-Instruct-FP8       | 81.98 ± 0.08 | 86.82 ± 0.97 | 37.67 ± 1.39 | 31.08 ± 1.96 |
| Mistral-7B-Instruct-v0.3     | 59.15 ± 0.10 | 35.90 ± 4.54 | 8.94 ± 1.24  | 21.60 ± 0.17 |
| Mistral-7B-Instruct-v0.3-FP8 | 58.77 ± 0.13 | 32.20 ± 4.02 | 7.68 ± 1.23  | 22.72 ± 0.19 |

## Empirical takeaway

For most models, FP8 scores are within reported error bars on MMLU/GSM8K, leaving a weak signal for black-box detection.

Data: paper benchmark table (original vs FP8 quantization)

# Higher Temperature Amplifies Performance Gaps but Increases Noise



Temperatures 0.2→2.0 degrade accuracy across tasks for all models.

At higher temperature, substitution signal can narrow or become dominated by sampling noise, complicating statistical tests.

# Log Probability Fingerprinting Fails Due to Inference Nondeterminism

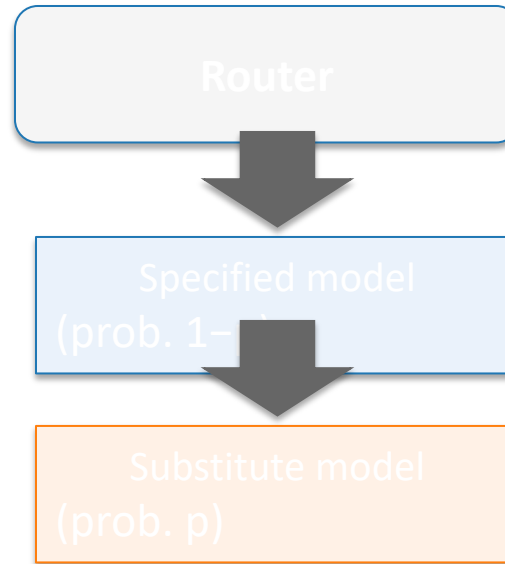
| Model                       | BERT Acc | T5 Acc | GPT2 Acc | LLM2Vec Acc |
|-----------------------------|----------|--------|----------|-------------|
| Llama3-70B-Instruct-FP8     | 50.55    | 50.10  | 50.45    | 49.90       |
| Llama3-70B-Instruct-INT8    | 51.60    | 49.90  | 51.30    | 50.25       |
| Gemma2-9b-it-FP8            | 49.95    | 50.30  | 51.20    | 49.80       |
| Gemma2-9b-it-INT8           | 49.00    | 49.70  | 51.65    | 49.55       |
| Mistral-7b-v3-Instruct-FP8  | 50.55    | 49.75  | 48.70    | 48.75       |
| Mistral-7b-v3-Instruct-INT8 | 49.70    | 50.75  | 50.50    | 51.15       |
| Qwen2-72B-Instruct-FP8      | 50.05    | 50.25  | 48.75    | 49.80       |
| Qwen2-72B-Instruct-INT8     | 50.75    | 50.55  | 49.45    | 50.20       |

Same model yields different token log-probabilities across software stacks (transformers, vLLM) and GPUs (A100 vs H100). Auditors cannot separate infrastructure drift from true substitution — fingerprints are non-robust in production.

Figure: log-probability variation across frameworks/hardware (figure\_5.jpg)

# Randomized Routing at Low Rates Makes Statistical Detection Query- Prohibitive

- Attack: route each query to substitute with probability  $p$ .
- Observed distribution:  $P_{\text{mix}} = (1-p) \cdot P_{\text{spec}} + p \cdot P_{\text{sub}}$ .
- As  $p \rightarrow 0$ ,  $P_{\text{mix}} \rightarrow P_{\text{spec}}$  — distinguishing becomes increasingly query-intensive.
- Providers can also combine with benchmark evasion (route known eval prompts to genuine model).



# Software-Only Auditing Methods Are Fundamentally Unreliable

| Method type                               | Core weakness                                                        | Practical limitation                                                                                                |
|-------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Text-output statistical tests (e.g., MMD) | Subtle substitutions (e.g., FP8) shift output distribution weakly    | Query budgets become impractical; randomized routing drives $p \rightarrow 0$ and makes detection query-prohibitive |
| Log-probability fingerprinting            | Production inference is non-deterministic across frameworks/hardware | Cannot separate infra drift from substitution; fingerprints are not stable                                          |
| Adversarial countermeasures               | Provider can adaptively route/evasive-handle audit prompts           | Benchmark evasion and randomized substitution undermine any fixed audit protocol                                    |

## Bottom line

Software-only audits lack robustness against (i) small accuracy deltas, (ii) real-world nondeterminism, and (iii) adaptive providers.

# Trusted Execution Environments

## Provide Provable Cryptographic Model Integrity

- TEE enclaves provide cryptographic attestation of code + data executed.
- Provider can prove specific model weights are loaded and correct inference code is running.
- Practical today (e.g., NVIDIA confidential computing on H100) with modest overhead.
- Contrasts with ZK proofs: currently computationally prohibitive for large models.

### Provable

Hardware-rooted attestation binds model+runtime to a measurement

### Robust

Unaffected by sampling temperature, randomized routing, or nondeterminism

### Actionable

Deployable in current GPU clouds with existing confidential computing



# Key Takeaways: Close the Verification Gap with Hardware Attestation

- Model substitution is economically motivated and hard to audit in black-box APIs.
- FP8 quantization barely changes benchmark scores — detection signal is small.
- Inference nondeterminism breaks log-probability fingerprinting in real deployments.
- Randomized substitution + benchmark evasion make statistical tests query-prohibitive.
- Trusted Execution Environments provide the most actionable path to provable integrity today.

**Call to action: demand attestable LLM APIs (TEE-backed inference) as a baseline trust primitive.**